Supplementary Material
Verify, Look Again, Remember: Fail-Closed Vision-Language Checks, Active Look-Around and Safety-Monotone Mission Memory for Relief-Supply Delivery Robots

Anonymous Authors

Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

This document holds material moved out of the main paper. Section, table and figure numbers with an S prefix refer to this document; unprefixed references are to the main paper. All numbers are generated from the same result files as the main paper.

S1. Failure Analysis and Lessons

Running the full stack on a GPU revealed problems that 606 offline unit tests and mocked ROS runs had not.

- **The auditor trusted the controller's sensor.** The wheel joint axis was expressed in the child-link frame after a 90° roll, so the wheels spun about the vertical axis. Odometry reported 1.5 m forward motion while the body slid sideways by 0.75 m. An audit based on `/odom` would have passed the run. Audits now use Gazebo's ground-truth odometry. Changing the wheel collision from a cylinder to a sphere reduced dead-reckoning drift from 0.16 to 0.011 m.
- **Free-form JSON from the first VLM was unusable** on our renders. With Qwen2.5-VL-3B, 21 of 48 outputs could not be parsed and the model hallucinated water on empty shelves. A newer model parses reliably, but it still approves fewer correct deliveries than scoring does (Table II). This motivated closed-set scoring. T1 measures the free-JSON method again (Table II).
- **Middleware effects.** Best-effort QoS silently dropped 92 of 102 0.9 MB camera frames under load. The GPU lidar detected the robot's own camera mast. Slow DDS discovery on a 4-core host failed 4 of 12 starts in an aborted campaign. An interrupted supervisor left an orphaned stack that made every trial of the next campaign stop with two `/cmd_vel` publishers. All of these are recorded with their fixes; the aborted campaigns are published.
- **Stale-request latch.** Two pre-decision retries in campaign D happened while the robot was stationary: the mission node's request stream stalled for ≈0.3 s on the saturated CPU. We instrumented the mission loop, the guard and garbage collection and rejected two hypotheses (guard starvation, GC pauses). Raising control-loop priority did not remove the stalls. We therefore changed the guard: a stale zero request now holds zero, while a stale motion request still latches. With a 0.6 s stall injected before departure, the old guard failed 2/2 runs and the new one completed 2/2. Raising priority with a file capability then had a side effect: it put the exec into the dynamic loader's secure mode, which stripped `LD_LIBRARY_PATH` and broke the ROS bag recorder. A regression run caught this before any campaign.
- **A pilot exposed a threshold that was not conservative enough.** Before campaign E we ran nine trials in separate pilot worlds. In one, the robot found the wrong item, searched the other station, and delivered from it although a fire burned behind it. The fire was almost entirely hidden by the first-aid kit, and $p_h = 0.32$. On development data no fire-free image exceeded $p_h = 0.0007$, so we lowered τ_h to 0.05 (v3.5). The same pilot worlds then gave 9/9 expected outcomes. T1 was not involved: both thresholds were registered for T1 before any T1 image was scored.
- **A generator bug caught by looking at the data.** In the first T1 render, distractors meant to stand next to a supply were placed through it. The emptiness check ran after the supplies were moved. The render was discarded and the fix recorded as a pre-registration amendment, before any model scored a T1 image. A second amendment replaced the field-of-view fire label with the rendered-mask label, after we saw fires hidden behind supplies that the geometric label had called visible.

S2. Wall-clock timing

Where the time goes. Table S1 breaks down the 34 delivered missions that needed no search. The check itself takes about 5.5 s: aiming, waiting until stationary, frame checks, and the scored VLM queries (3.4 s). Driving between station and shelter takes about 39 s (sum of phase medians). The check is therefore short compared with driving. The situation report adds about 4 s but could run in parallel with the drive to the shelter.

TABLE S1
WALL-CLOCK PHASES OF DELIVERED MISSIONS WITHOUT SEARCH (CAMPAIGNS D AND E; V3.4 AND V3.5, NO LOOK-AROUND).

Phase	Median [s]	Min	Max
drive to station	14.31	12.19	16.62
aim camera	0.50	0.50	0.52
wait until stationary	0.67	0.66	0.71
acquire and check frames	0.90	0.75	0.98
VLM decision (scored queries)	3.44	2.95	6.64
situation report (not used to decide)	5.14	3.99	8.54
drive to shelter	24.83	15.02	25.32

TABLE S2
DECISION-RULE ABLATIONS ON T1: FALSE APPROVALS / FALSE DENIALS. *: EXACT MCNEMAR $p < 0.05$ VERSUS THE FULL RULE OF THE SAME MODEL.

Decision rule	Qwen3.5-9B	Qwen3.5-4B	Qwen3-VL-8B	Qwen2.5-VL-3B
Full rule	5 / 4	9 / 1	4 / 8	4 / 4
A1: question 1 only	5 / 4	9 / 0	4 / 7	4 / 4
A2: question 2 only	5 / 1	9 / 1	4 / 2*	36 / 1*
A3: no hazard question	47 / 4*	48 / 1*	46 / 8*	48 / 4*
A4: no threshold (τ_s=0)	5 / 4	9 / 1	4 / 8	4 / 4
v3.5: τ_h=0.05 (I-10)	3 / 4	4 / 1	4 / 8	1 / 30*

S3. ADDITIONAL T1 ANALYSES

A. Decision-rule ablations

Table S2 isolates the parts of the rule on T1 (Table II, Eq. 1). For the 9B model neither the permuted second question nor the confidence threshold changed the number of false approvals. With Qwen2.5-VL-3B, requiring the two multiple-choice questions to agree reduced false approvals, which is the option-letter bias of [1].

B. Risk–coverage

Fig. S1 varies τ_s only for description, because τ_s was not tuned on T1.

C. Weaker models and free-form output

For Qwen2.5-VL-3B the second question alone approved 36 must-refuse images, and requiring agreement with the first question brought this down to 4, which is the option-letter bias of [1] in action (Table S2). Asking this model for free JSON (A5) left 148 of the 1200 generations unparseable and approved 57 images; Qwen3.5-9B gave no parse errors (main paper, Sec. V).

S4. CAMPAIGN E DETAILS

Campaign E used a new randomised world in every trial (main paper, Sec. V); Table S3 gives the outcomes by stratum and Fig. S2 the ground-truth trajectories.

S5. MODEL SELECTION ON DEVELOPMENT DATA

Models were ranked lexicographically. The first key was zero false approvals and zero unsafe intent acceptances, and the second was explanation and intent accuracy. Qwen3-VL-8B parsed more development utterances but accepted two unsafe ones. Qwen3.5-4B produced a false approval. Qwen3.5-9B was therefore chosen and frozen before the T1 and E registrations.

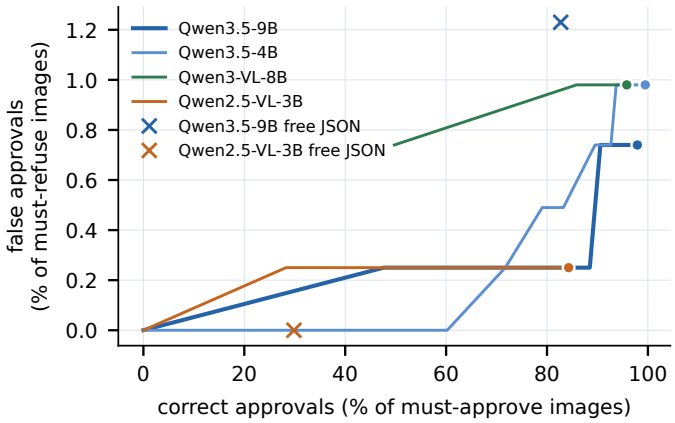


Fig. S1. T1 risk–coverage: false-approval rate versus correct-approval rate as τ_s varies ($\tau_h = 0.05$, v3.5). Markers show the frozen $\tau_s = 0.5$. The curves are flat over most of the range: the remaining errors are missed fires, which τ_s cannot fix.

TABLE S3
CAMPAIGN E (RANDOMISED WORLDS) BY STRATUM.

Expected	Condition	n	Correct	Observed
allow	box	2	2	allow 2
allow	cone	4	4	allow 4
allow	crate	2	2	allow 2
allow	plain	12	12	allow 12
deliver after search	box	2	2	allow after search 2
deliver after search	crate	2	2	allow after search 2
deliver after search	plain	12	12	allow after search 12
deny	box	2	2	deny 2
deny	crate	4	4	deny 4
deny	plain	6	6	deny 6
deny hazard	fire	8	8	deny hazard 8
deny hazard after search	fire other	4	4	deny hazard after search 4

S6. REGISTRATION AUDIT

For every study with results, the registration commit precedes the results commit (holds). Campaign C's registration and ledger entered the repository together, so their order cannot be verified from the history.

S7. EXTENDED RELATED WORK

The main paper condenses this section; the full text follows.

Failure detection and verification with foundation models. REFLECT [2] and AHA [3] explain manipulation failures in natural language after the fact. Inner Monologue [4] feeds success detection back into a planner. Du et al. [5] fine-tune a VLM as a success detector. Sentinel [6], FAIL-Detect [7] and Code-as-Monitor [8] monitor learned policies at run time, an LLM-based anomaly monitor triggers reactive fallbacks [9], and uncertainty-aware policy steering decides when to act, ask or learn [10]. Semantic anomaly detection with LLMs [11] and visual question answering that abstains rather than answers wrongly [12] are close in spirit to a gate that prefers refusal. Our check runs *before* the robot acts and is closed-set and zero-shot. Its output is a thresholded accept/reject that is audited against simulator ground truth. The closest methodological relative is KnowNo [13], which scores multiple-choice answers by next-token probability and asks a human for help,

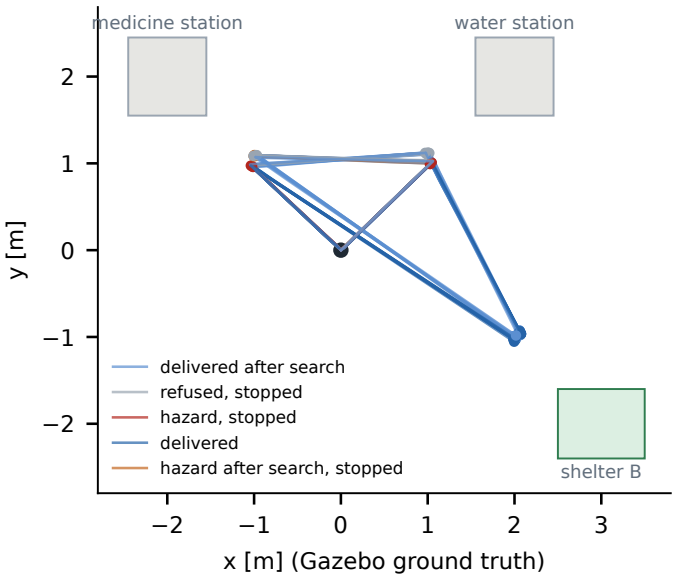


Fig. S2. Campaign E: ground-truth trajectories of all final trials, coloured by audited outcome.

TABLE S4
CANDIDATE MODELS WITH THE FINAL PROMPTS (`v35`: CHECKPOINT QUESTIONS OF V3.5; `i35`: INSTRUCTION PARSER) ON DEVELOPMENT DATA ONLY. IMAGES: CORRECT DECISIONS WITH FALSE APPROVALS (FA) AND FALSE DENIALS (FD). EXPLANATIONS: ONE-SENTENCE REPORTS JUDGED CORRECT AGAINST THE GENERATOR LABELS. INTENT: DEVELOPMENT UTTERANCES, WITH UNSAFE ACCEPTANCES. VRAM IN MIB.

Model	Released	Images (176)	Explanations (128)	Intent dev (40)	VRAM
Qwen2.5-VL-3B	2025-01	176/176 (FA 0/FD 0)	84/128	31/40, unsafe 0	–
Qwen3-VL-8B	2025-10	176/176 (FA 0/FD 0)	120/128	38/40, unsafe 2	17212
Qwen3.5-4B	2026-02	175/176 (FA 1/FD 0)	125/128	35/40, unsafe 2	9102
Qwen3.5-9B	2026-02	176/176 (FA 0/FD 0)	125/128	35/40, unsafe 0	18320

with a conformal-prediction coverage guarantee (see [14] for background on conformal prediction). SayCan [15] likewise scores a fixed candidate set by language-model likelihood. We share the scoring, but we respond to uncertainty by refusing or searching, and we add permuted-option agreement as a test-time defence against letter bias.

Multiple-choice scoring and its biases. Scoring answer symbols works well when a model binds symbols to options [16]. Language models are reasonably calibrated on well-formatted multiple-choice questions [17], but they prefer certain option IDs [1] and orders [18], and their raw confidence needs validation [19], [20]. Our two permuted questions are a minimal, deterministic form of self-consistency [21]. The Yes/No hazard question follows the polling style of POPE [22].

Safety architectures. Simplex [23], [24], shields [25] and control barrier functions [26] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [27], and guardrails such as RoboGuard [28] check plans before execution. In contrast to vision-language-action models [29], [30], our VLM never commands motion.

TABLE S5
WHEN EACH STUDY WAS REGISTERED AND WHEN ITS FIRST RESULTS ENTERED THE REPOSITORY. "ORDER" CHECKS THAT THE REGISTRATION COMMIT IS AN ANCESTOR OF THE FIRST RESULTS COMMIT (`GIT MERGE-BASE --IS-ANCESTOR`). REGISTRATIONS ARE ALSO MIRRORED IN THE PUBLIC REPOSITORY HISTORY.

Study	Registered (commit, date)	First results (commit, date)	Amendments	Order
C (v3.3)	`5c9a951 2026-10-16`	`5c9a951 2026-10-16`	0	same commit
D (v3.4)	`ac6d815 2026-10-16`	`fb4e9b3 2026-10-16`	0	✓
T1	`5f835af 2026-10-16`	`c3d1ffc 2026-10-16`	3	✓
E	`6201aa0 2026-10-16`	`17e0ea6 2026-10-16`	0	✓
T2	`f347239 2026-10-17`	`7843e9c 2026-10-17`	1	✓
F	`564a724 2026-10-17`	`5bbf11a 2026-10-17`	0	✓
G	`edb26a1 2026-10-17`	`f6160fb 2026-10-17`	1	✓
R	`fb63bee 2026-10-17`	`2337e49 2026-10-17`	0	✓
H	`a5cc651 2026-10-17`	`pending`	1	pending
J	`b72b877 2026-10-17`	`pending`	0	pending
Z	`a53ee7e 2026-10-17`	`0ca401d 2026-10-17`	0	✓
P	`e6bde02 2026-10-17`	`00795a8 2026-10-17`	0	✓

Memory for robots in changing environments. Robots increasingly keep long-horizon semantic memory. ReMEmbR [31] and Embodied-RAG [32] retrieve from a database of captioned observations; KARMA [33] adds long- and short-term memory to embodied agents, and H-EMV [34] verbalises a robot's episodic memory. 3D scene graphs such as Hydra [35], ConceptGraphs [36] and HOV-SG [37] support planning [38] and language grounding [39]. Long-term mapping has long dealt with change [40], [41], and recent work tracks changes in dynamic scene graphs [42], [43]. These systems use memory to *answer* or *plan*. A recent preprint reports, in a small grid-navigation testbed, that acting on unaudited stale memory can be worse than having no memory [44]. GLOVE [45] and VLM-DEWM [46] check memory or a world model against fresh observations to keep the memory *correct*. We study a different use: memory decides *where and how carefully to verify*. We restrict it so that it can never reduce verification, and we state this as a provable invariant and test it under scripted change.

Active perception. Moving the sensor to resolve ambiguity is a classic idea [47], [48], and view planning for inspection is well studied [49]. Recent embodied agents explore until a VLM is confident [50] or let a VLM choose views [51]. Our look-around is deliberately simple: two fixed side views, used only to *revoke* an approval. This makes its effect auditable and its safety direction obvious.

Robustness and the sim-to-real gap. Image classifiers degrade under common corruptions [52], and so do detectors [53], adapted vision-language models [54] and video-language models [55]. Sim-to-real transfer is a recognised challenge in robotics [56]. Without a physical platform, we measure one part of this gap, sensor degradation, under a pre-registered protocol. What we care about is the *direction* of the errors, not only accuracy.

Diverse redundancy. Requiring agreement between independently developed versions is a classic fault-tolerance technique [57]. Its benefit is limited by correlated failures [58], and model-agreement schemes for language models have reappeared recently [59]. We report both sides: the two models agree on most images, and the second model's value depends strongly on how it is asked.

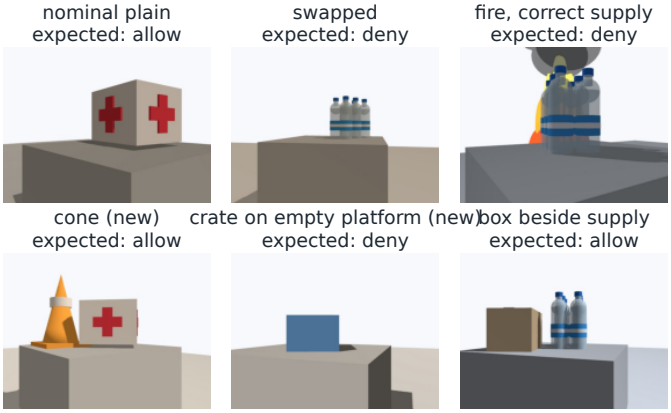


Fig. S3. Held-out T1 images (robot camera, 640×480) with the generator's expected decision. The crate and the traffic cone never appeared during development.

Evaluation practice. We follow recommendations to report conditions, multiple metrics, uncertainty and failure modes [60]. We use Wilson intervals [61] and exact McNemar tests [62]. Domain randomisation [63] is used here to build a held-out *test* distribution, not for training.

S8. Development Data, T1 Construction and the Two-Model Rule

Development data. Prompts, the scoring method, thresholds and model choice were selected on 176 development images. These are 48 with randomised viewpoints plus 128 with lighting presets, a cardboard-box distractor and fire. The instruction parser was tuned on 40 development utterances. The later hazard fixes (main paper, Sec. V-C) also used two hazard development sets of 300 rendered images each: DH, and DH2, which emphasises small early-stage fires.

Two-model rule. The two-model rule (main paper, Sec. III-C) approves only if Eq. (1) holds for Qwen *and* CLIP-Flash names the requested supply with confidence ≥ 0.65, presence ≥ 0.80 and hazard probability below its own threshold.

T1 examples. Fig. S3 shows held-out T1 images. T1 used 60 Gazebo worlds × 2 stations × 5 viewpoints with continuous random lighting, supplies moved by up to ±0.15 m and freely rotated, random fire positions, and three distractors. Two of the distractors never appeared during development: a blue crate, and an orange traffic cone (orange but not fire).

S9. Further Result Details

Study Z. The contrastive baselines used SigLIP [64] and CLIP [65] prompts for the supply and the hazard, and the same crops as our check. On development data no threshold pair gave both few false approvals and many correct ones. The contrastive scores do not separate "right supply, no fire" from the alternatives in these scenes.

T1 hazard-question false alarms. With the registered v3.4 rule the hazard question raised false alarms on 0/25 traffic-cone images and on 0/523 images without fire. With the v3.5 rule, it raised false alarms on 0/25 cone and 0/523 fire-free images.

TABLE S6
Study R: v3.6 offline check under camera degradations (T2 subset). Severity M = moderate, S = severe.

Degradation	Sev.	False appr.	Fire missed	Correct appr.	Gate rej.
None (clean)	–	0/192	0/27	107/108	0
Sensor noise	M	0/192	0/27	107/108	0
Sensor noise	S	0/192	0/27	107/108	0
Motion blur	M	0/192	0/27	106/108	0
Motion blur	S	0/192	0/27	101/108	0
Defocus	M	0/192	0/27	108/108	0
Defocus	S	0/192	0/27	108/108	0
Under-exposure	M	0/192	0/27	107/108	0
Under-exposure	S	0/192	0/27	106/108	0
Over-exposure	M	0/192	0/27	96/108	4
Over-exposure	S	0/192	0/27	59/108	24
JPEG	M	0/192	0/27	108/108	0
JPEG	S	0/192	0/27	107/108	0

Campaign F, remaining misses. Six of v3.6's seven remaining misses were fires of 1 100–2 400 visible pixels with hazard probabilities of 0.002–0.006, which overlap those of fire-free views, so no threshold can separate them. The seventh showed 8 600 pixels at $p_h = 0.019$.

Campaign G, world set and medians. The set of worlds is defined from mask renders at all three views, so it includes the side views by construction. The medians over all delivered runs, 102 s and 122 s, are not comparable, because look-around refused most of the fire runs that the front-only system delivered.

Study P, false alarms and licence. D-Fire [66] was released by its authors under CC0, as stated in the original repository's LICENSE; we used a Hugging Face mirror. Rendered fire is stylised, which motivated this check on real photographs. D-Fire's fire-free class comes from fire-surveillance cameras and is rich in hard negatives: hazy hillsides, low clouds, a smudge on the lens, red sunset clouds, glowing tents. Everyday scenes such as streets, beaches and roads scored $p_h \approx 0.01$. The question was worded to catch faint smoke ("a grey or black cloud") and thresholded at 0.05, so it errs strongly toward stopping on real images. We registered the site-specific threshold calibration as a follow-up study (P2).

Campaign H, failure types. Staleness predicts two failure types for naive memory: delivering a supply that has been moved away, and delivering where a fire has appeared since the last visit; the per-trial ledger records which occurred.

Study R, all conditions. Table S6 gives the false approvals, correct approvals and acquisition-gate rejections of study R (main paper, Sec. V-D) for every degradation and severity.

Hazard fixes on development data. Table S7 compares the candidate hazard fixes of the main paper (Sec. V-C).

Campaign H, per-mission outcomes. Fig. S4 shows every mission of campaign H (main paper, Sec. V-E).

Repeated-mission model. Fig. S5 extends campaign H (main paper, Sec. V-E).

TABLE S7
HAZARD FIXES ON DEVELOPMENT DATA (DEV V1+V2, DH, DH2). $\tau_h = 0.05$ UNLESS STATED.

Configuration	False appr.	Correct appr.	Fires refused	One V100
v3.5: Qwen3.5-9B	46/413	357/360	165/211	yes
+ enlarged crops	37/413	356/360	174/211	yes
+ hidden-fire wording	25/413	357/360	186/211	yes
+ wording + crops (v3.6)	17/413	357/360	194/211	yes
Qwen3.5-4B	37/413	360/360	174/211	yes
Clef-Flash, original Jev request	45/413	182/360	166/211	yes
Clef-Flash, task wording	42/413	350/360	169/211	yes
Clef (task) + Qwen (v3.6), both approve	15/413	350/360	196/211	no

TBD: fig_h_outcomes.pdf

Fig. S4. Campaign H, missions 2 and 3 of the 16 sequences. (a) Outcome of each mission under no memory (M0), naive memory (M1) and verified memory (M2); rows are sequences, grouped by scripted story (S1–S4). A failed mission is hatched. (b) Mission time of the missions whose relocation persisted, for each arm; lines join the M0 and M2 times of the same mission, bars mark medians.

TBD: fig_h_change_rate.pdf

Fig. S5. Model of repeated missions with the per-check rates measured in G and H ($p_{\text{det}} =$ **TBD**, $n =$ **TBD**): unsafe deliveries and mean mission time of the three memory policies as the world changes more often. The curves are computed from measured per-check rates, so their order is a model prediction, not an observation.

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